



NAME:.....INDEX NO.:.....

SCHOOL:.....CANDIDATES SIGNATURE:.....

DATE:.....

231/1  
BIOLOGY  
PAPER 1.  
THEORY  
TIME:

*Kenya certificate of secondary education (K.C.S.E)*

231/1  
BIOLOGY  
PAPER 1.  
THEORY  
TIME:

**Instructions to candidates.**

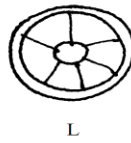
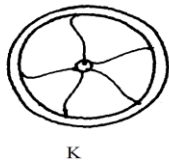
- a) Write your name and index number in the spaces provided above.
- b) Sign and write the examination date .
- c) Answer all the questions in the spaces provided in the question paper.

*This paper consists 12 of printed pages  
Candidates should check to ascertain that all pages are printed as indicated and that  
no question is missing.*



1. State ways by which synaptic transmission can be stopped. (2mks)  
.....  
.....  
.....  
.....
2. State two advantages which the endothermic (homoethemic) have over those that are exothermic (poikilothermic). (2mks)  
.....  
.....  
.....  
.....
3. a) State one function of red blood cells. (1mk)  
.....  
.....  
b) Give two structural difference between red blood cells and white blood cells. (2mks)  
.....  
.....
4. Hairs and leaves of sundew, an insectivorous plant curl around and trap insects when they land on the plant.  
a) Identify the response shown sundew plants (1mk)  
.....  
.....  
b) Explain the biological importance of the response in (a) above. (2mks)  
.....  
.....  
.....  
.....  
c) Name any one type of neurons. (1mk)  
.....  
.....

5. The diagram below gives an external view of the structure of the human eye observed outdoor at midday and midnight.



- a) Which diagram represents the eye as observed during the day? (1mk)

.....

.....

- b) Give a reason for your answer in (a) above. (1mk)

.....

.....

6. A student viewed and drew a plant cell of a diameter 4mm using a light microscope whose eye piece lens was marked x1 and objective lens marked x5. How many cells were linearly arranged along the microscope's field of view whose diameter was 8mm .

( show your working ) (3mks)

.....

.....

.....

.....

.....

.....

.....

.....

7. Identify the nucleic and whose base sequence is shown below.

G-A-C-U-A-G-A-C-G

- i) Identify the type of nucleic acid as shown below (1mk)

.....

.....

- ii) Give reason for your answer in (i) above. (1mk)

.....

.....

- iii) Write the base sequence of the DNA strand shown above (1mk)

.....

.....

8. State the function of the following parts of a microscope

(3mks)

i) Nose piece.

ii) Condenser.

iii) Diaphragm.

9. Identify the mode of feeding of the animal whose dental formula is given below.

$$\begin{matrix} 0 & 0 & 3 & 3 \\ i & C & Pm & M \\ 3 & 1 & 3 & 3 \end{matrix} = 32.$$

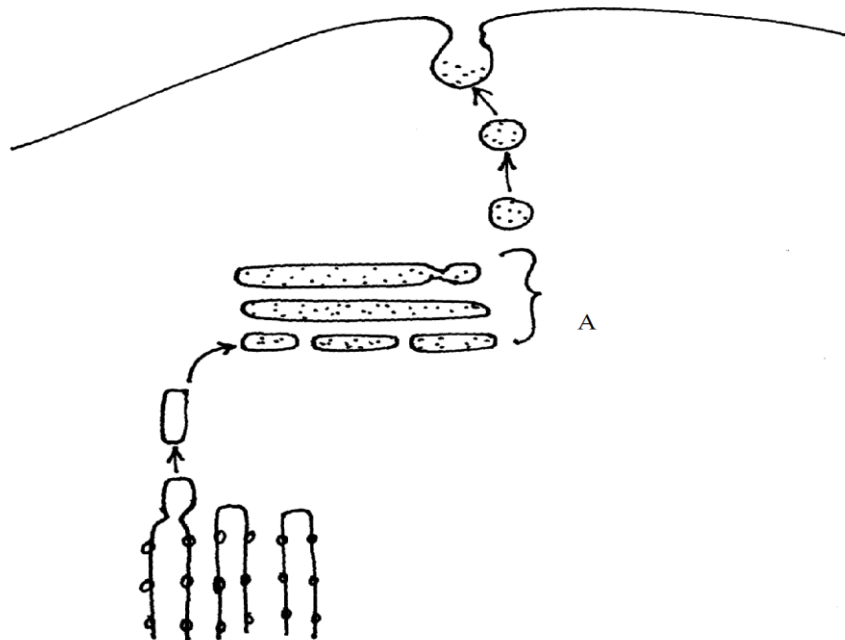
a) Mode of feeding

(1mk)

b) Give a reason for your answer in (a) above.

(2mks)

10. Study the diagram below and use it to answer the questions.



a) Identify the organelle marked A.

(1mk)

b) Give three functions of the organelle named in (a) above

(3mks)



.....  
.....  
.....  
.....  
11. It was found that during germination of pea seeds  $9.3\text{cm}^3$  of carbon (iv) oxide was produced while  $9.1\text{cm}^3$  of oxygen was used up.

a) Calculate the respiratory quotient (RQ) of the reaction taking place. (2mks)

.....  
.....  
.....  
.....

b) Identify the type of food substance being metabolized. (1mk)

.....  
.....

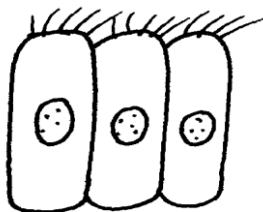
12. Explain why Lamarck's theory of evolution is not accepted by biologists today. (2mks)

.....  
.....  
.....  
.....

13. Give three reasons why plants lack complex excretory organs like those of animals. (3mks)

.....  
.....  
.....  
.....  
.....  
.....

14. The diagram below shows a type of epithelial tissue.



i) What is the name of the hair-like process? (1mk)

.....  
.....

ii) What is the function of the hair-like process. (1mk)

b) Name one part in the human body where the hair-like process are found. (1mk)

15. In an attempt to estimate the number of weaver birds in a small woodland 435 were captured , marked and released. Three days later , 620 were captured 75 of which were marked.

a) What is the name of the sampling method described above. (1mk)

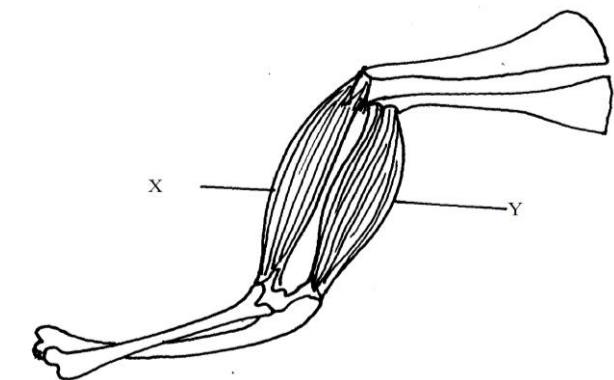
b) Calculate the approximate size of the weaver bird population in the woodland. (2mks)

c) Give one disadvantage of this method. (1mk)

16. Give an example of ball and socket joint. (1mk)

17. Name two types of strengthening tissues in plants. (2mks)

18. Study the diagram below, and answer the questions below.



a) Name the muscles labelled X and Y. (2mks)



b) What happens to each muscle as the arm is straightened? (2mks)

.....

.....

.....

19. Nocturnal animals such as a leopard are capable of seeing fairly well at night. What two retinal adaptations have made this possible? (2mks)

.....

.....

.....

20. The table below shows stomata distribution on leaves A and B and their surface area. Use the information to answer the questions that follow.

|                   | Leaf A            | B                 |
|-------------------|-------------------|-------------------|
| Number of stomata | Upper surface 25  | 5                 |
|                   | Lower surface 0   | 20                |
| Surface area.     | 30cm <sup>3</sup> | 19cm <sup>3</sup> |

b) Identify with reasons the habitats of the plant from which the leaves were obtained (4mks)

Leaf A      Habitat.....

              Reasons.....

.....

.....

Leaf B      Habitat.....

              Reasons.....

.....

.....

21. A tall bean plant crossed with a dwarf one produces offspring of which about half are tall and other are dwarf . what are the genotypes of parents? Show your working (3mks)

.....

.....

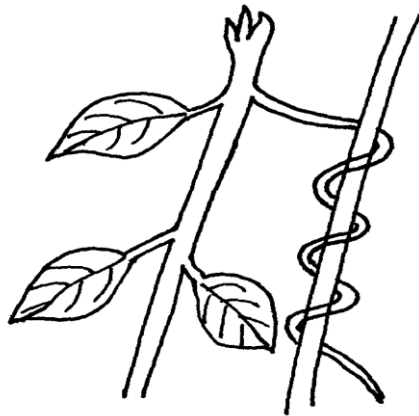
.....

.....

.....

22. Describe what happens during the dark stage of photosynthesis. (3mks)

23. The response exhibited by a certain plant tendril is illustrated below.



i) Name the type of response. (1mk)

ii) Explain how the response named in (i) above occurs . (2mks)

24. State three adaptations of respiratory surfaces. (3mks)

25. State the parts of the ear involved in:

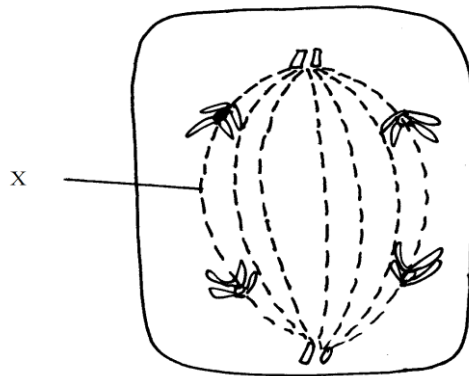
a) Amplification of sound vibration (1mk)

b) balance and posture. (1mk)



26. Explain why the digestion of starch stops after food enters the stomach. (2mks)

27. The diagram below represents a stage during cell division.



i) Identify the stage of cell division. (1mk)

ii) Give two reasons for your answer (a)i) above (2mks)

iii) Name the structure labelled M. (1mk)

28. Bivalent synapsis, crossing over are terminologies used in cell division.

a) Name the stage of meiosis in which the process takes place. (1mk)

b) Distinguish between synapsis and crossing over. (2mks)